上海交通大学
SHANGHAI JIAO TONG UNIVERSITY

Carbon Accounting

Lecture 4

Carbon Accounting: Attributional vs Consequential

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A New Use for Whisky Residues

Whisky production leaves grain residues that have traditionally been sold as animal feed.



Whisky distillery



Grain residues



Used as boiler fuel

replaces



Less fossil fuel

The distillery plans to burn the residues in its boiler, replacing some fossil fuel.

Pictures are generated by GPT-5.6

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设置

The distillery plans to burn its grain residues as fuel, replacing some fossil fuel. Do you think this change will reduce greenhouse gas emissions?

A Yes

B No

C Not sure

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What Else Might be Changed by the Plan?

Would lower fossil-fuel use at the distillery tell us the whole story?



Where did the residues go before?



What might replace them?



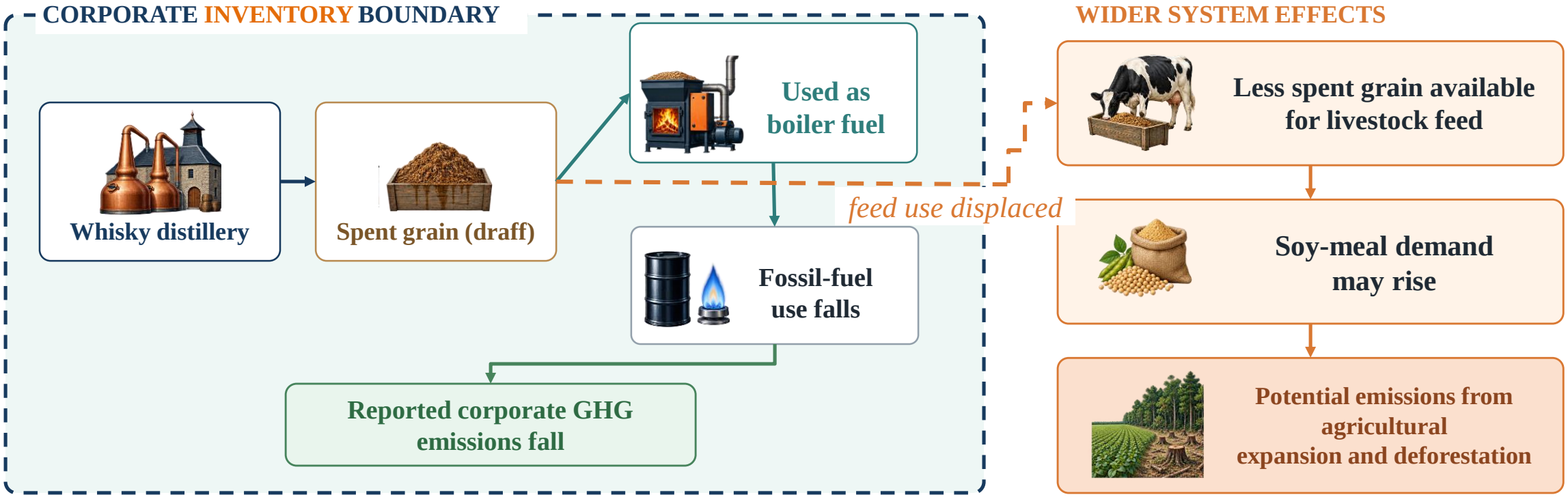
What other activities might change?

The case contains two different accounting questions.

Pictures are generated by GPT-5.6

Whisky Residues as Fuel

Emissions are reduced or not?



ATTRIBUTIONAL QUESTION

What emissions are assigned to the distillery?

The corporate inventory improves.

CONSEQUENTIAL QUESTION

What emissions change because of the decision?

The net effect cannot be read from the inventory alone.

Generated by GPT-5.6

Learning Outcomes

1. **Distinguish between** attributional and consequential carbon accounting based on the questions they answer.
2. **Explain** how boundary rules shape attributional results and how baselines shape consequential estimates.
3. **Apply** the common analytical framework to determine which accounting approach fits a given purpose and intended use.

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01

Attributional and Consequential Accounting

Two approaches for two different questions

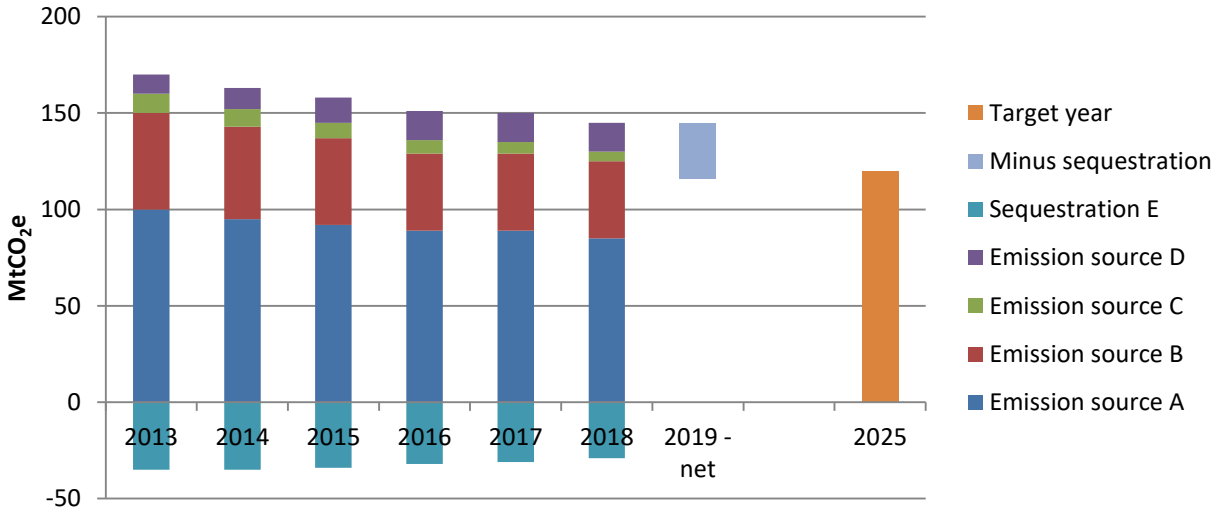
Two Types of Accounting Questions

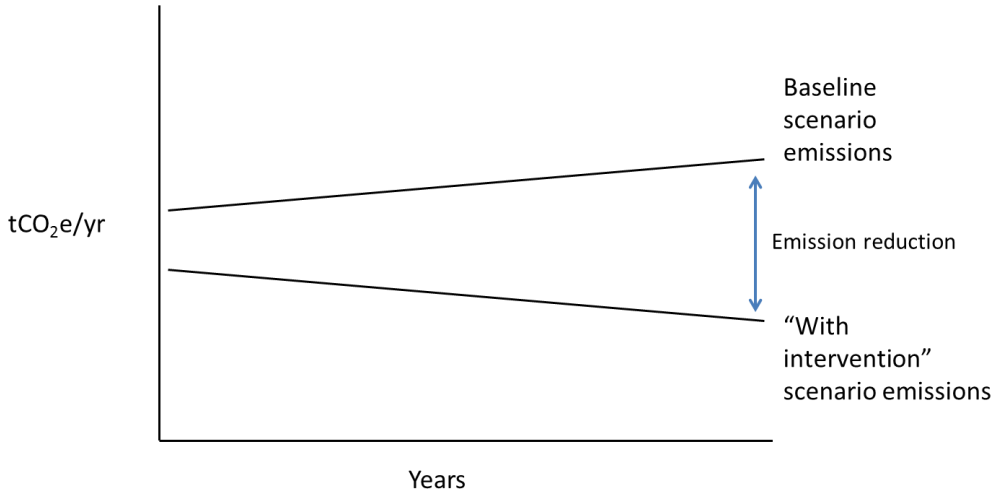
- Carbon accounting may ask two different types of questions:

  How much emissions are associated with/allocated to this object?

  How do emissions change because of this decision?

- These two questions lead to different accounting approaches.





Attributional Carbon Accounting

Attributional (Allocational) Carbon Accounting

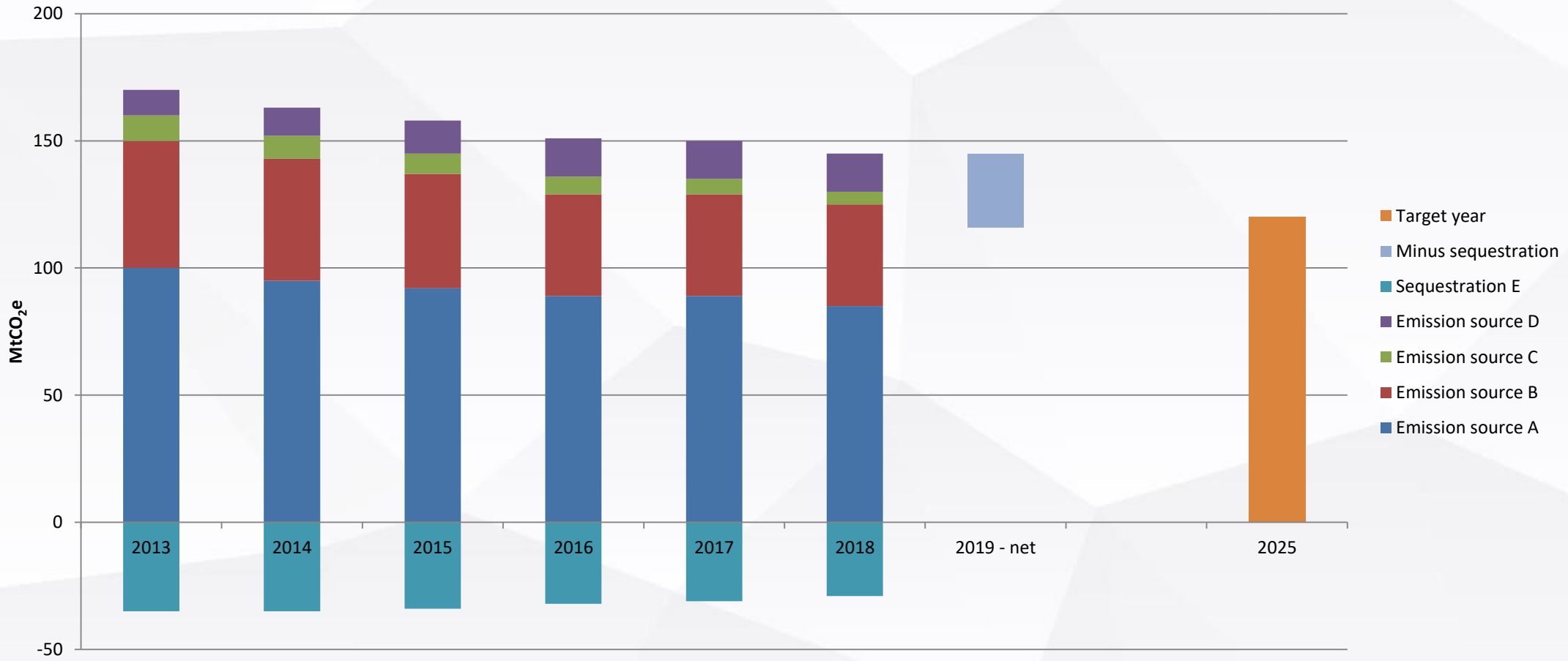
An inventory of anthropogenic GHG emissions and removals within a defined boundary.



- Typical features:
  - Defined object
  - Defined boundary
  - Activity data × emission factors
  - Often used for reporting, disclosure, compliance, and tracking
- Examples:
  - National GHG inventory
  - Corporate annual emissions
  - Product carbon footprint

Core question: What emissions should be counted within this boundary?

A schematic diagram of attributional carbon accounting



Whisky production: where do emissions come from?

Look closely at the production process before we decide what to count



Follow the production process

1. **Malt & mash** — grain, water, hot water
2. **Ferment** — yeast, cooling, time
3. **Distill** — boiler heat, steam, cooling
4. **Mature & pack** — barrels, glass, transport

Look for energy use and emission sources



Malt & mash → Ferment → Distill → Mature & pack

Questions

- Which steps use energy?
- What types of energy are used?
- How is energy use connected to greenhouse gas emissions?

What Should Count for SJTU?

SJTU as an organization



SJTU
Annual organizational emissions

- **Electricity used in university buildings**
- **Electricity used in a university-affiliated hospital**
- **Electricity used in student dormitories**
- **Staff and student commuting**
- **University-funded business travel**
- **Fuel used by university vehicles**
- **Production of laboratory equipment**
- **Methane released from university wastewater treatment**
- **Food served in campus canteens**
- **…**

Would the boundary change if the result were used to decide whether to approve a campus electric-shuttle project?

Flexible Boundaries Need Shared Rules

If one account includes a source and another leaves it out, their results may not mean the same thing.

GUIDELINES HELP WITH

- Common categories
- Minimum requirements
- Consistent reporting

APPLICATION STILL REQUIRES

- Interpreting high-level rules
- Handling ambiguous sources
- Documenting choices

Shared rules improve consistency. Judgment remains part of application.

Consequential Carbon Accounting

A Different Accounting Question



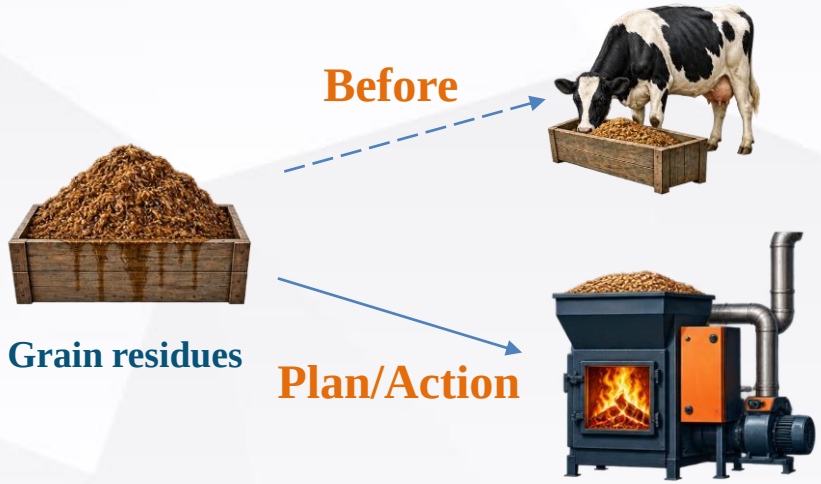
This question follows the effects of an action beyond one account.

Now change the question

What changes because the distillery switches fuels?

How much is counted for the distillery?





What Changes Because of the Fuel Switch?



Less fossil fuel



Less grain residue for animal feed



More alternative feed



Possible changes in land use

A **consequential** analysis follows activities that change because of the action.

Compared with What?

What would probably have happened without the fuel switch?

WITHOUT THE SWITCH

Fossil fuel use continues

Residues remain available as feed

No additional replacement feed

WITH THE SWITCH

Less fossil fuel

Residues burned as fuel

Farmers buy replacement feed

Counterfactual baseline

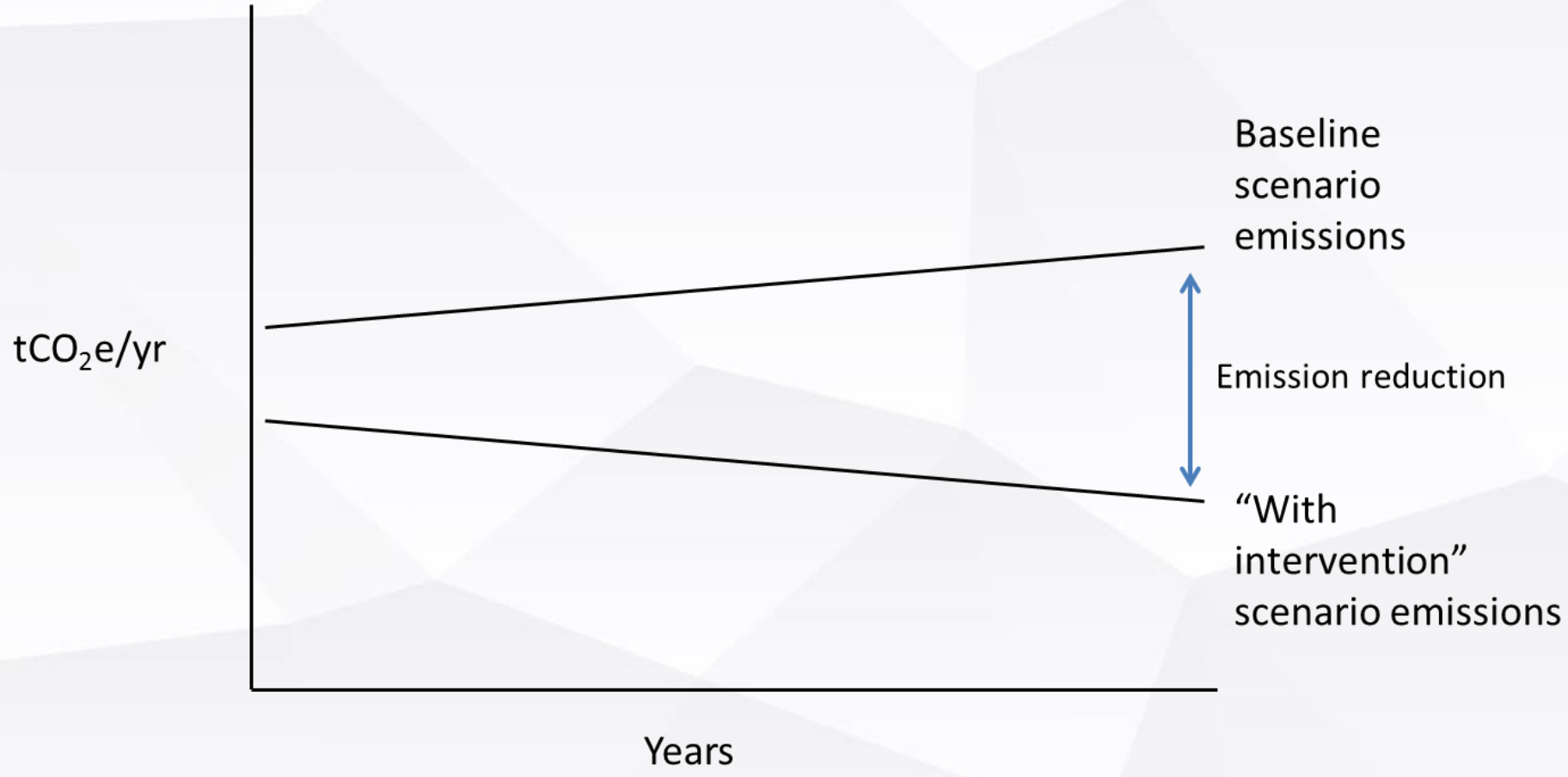
what would otherwise have happened

Consequential Carbon Accounting

Assessment of how GHG emissions change because of a specified action or intervention

- Typical features:
  - Baseline scenario
  - Intervention scenario
  - Indirect effects
  - Marginal changes
  - Often used for project or policy evaluation
- Examples:
  - Avoided emissions from a project
  - Policy impact assessment
  - Biofuel and land-use change

A schematic diagram of consequential carbon accounting



Attributional and Consequential Methods

They answer different questions, so they use different boundaries and comparison bases.

Feature	Attributional methods	Consequential methods
Purpose	Describe or allocate emissions within a defined account.	Estimate the system-wide change caused by a decision or intervention.
Boundary	Rules determine which sources and sinks belong to the account.	System affected by a decision or intervention.
Comparison basis	A reporting period. Trends compare with periods or a base year.	The intervention scenario compared with a counterfactual baseline.
Output	Emissions assigned to an entity, product, or period.	Estimated change in emissions relative to what would otherwise happen.
Typical use	Inventories, footprints, target tracking, and reporting.	Decision support and impact assessment.

For a full table, please refer to Brander (2021)

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Do you think bioenergy is a carbon-reducing energy source?

A Yes.

B No.

C It depends.

提交

02

A Common Analytical Framework

From framing the task to evaluating the result

Common Analytical Framework

Trace how a carbon accounting result was framed, modeled, quantified, and used

1

Framing the Accounting Task

Purpose and intended use

2

Examining the Accounting Model

Accounting subject
Boundary
Attribution

3

Quantifying Emissions

Accounting approach
Data and calculation

4

Evaluating and Using the Result

Quality and uncertainty
Interpretation and use

FITNESS FOR PURPOSE

Is the accounting result fit for its intended use?

Compare the intended use with what the result can credibly support.

Questions to ask when analyzing/framing a carbon accounting practice

Purposes and Intended Use	• Why is the accounting conducted? • Who will use the result, and for what? • What is the result intended to support?	**Why**
Object, Boundary and Attribution	• Whose or what emissions are being accounted for? • What is included—and excluded? • On what basis are emissions assigned to this account?	**What**
Data, Methods and Assumptions	• What evidence is used? • How is it converted into emissions? • What assumptions affect the quantification?	**How**
Quality, Meaning and Use	• What does the result show—and not show? • How credible and comparable is it? • What can it actually support, and what follows from its use?	**So what**

Overarching question: Is the result fit for its intended use?

Wrap up

What we have built today

1. **Two accounting approaches answer different questions:**

   attributional accounting describes an account; consequential accounting estimates a change.

2. **Attributional accounting:**

   Describes emissions associated with a defined entity or system, using stated boundaries and attribution rules. It supports inventories, reporting and tracking over time.

3. **Consequential accounting:**

   Estimates how emissions would change because of a decision or intervention, compared with what would otherwise happen—not simply the difference between two inventories.

4. **A common analytical framework:**

   WHY—purpose and intended use; **WHAT**—object, boundary and attribution; **HOW**—data, methods and assumptions; **SO WHAT**—quality, meaning and use.

Next Time – National-level Carbon Accounting

- Essential reading:

  IPCC (2006), 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 1, Chapter 1, pp. 1.4–1.8.(The whole guidelines is a vast document, there is a Chinese version too.)

  - Read Sections 1.1–1.4, approximately five pages. Focus on national territory, inventory year and time series, sectors and categories, activity data × emission factor, tiers, and inventory quality. You may skip the footnotes and details on additional gases.

- Recommended reading

  Guan, D., Liu, Z., Geng, Y. et al(2012). The gigatonne gap in China's carbon dioxide inventories. Nature Clim Change 2, 672–675. https://doi.org/10.1038/nclimate1560

- Further reading:

  Ritchie, H. (2019), "How do CO_2 emissions compare when we adjust for trade?", Our World in Data. https://ourworldindata.org/consumption-based-co2

Quiz (see Canvas)



In-class Course Project Activities

Prototype 0: your first prompts and your first “product”





See you next time